12 (a) Express $\frac{x}{x+2} + \frac{2x}{x-4}$ as a single fraction in its simplest form.

Common denominator: $(x + 2)(x - 4)$.

Numerator = $x(x - 4) + 2x(x + 2)$

$= x^2 - 4x + 2x^2 + 4x = 3x^2$.

Result = $3x^2 / [(x + 2)(x - 4)]$.

3 marks total
Common denominator
Combine numerators
$3x^2$ / product

.....

(3)

(b) Expand and simplify $(x - 3)(2x + 3)(4x + 5)$

$(x - 3)(2x + 3) = 2x^2 - 3x - 9$.

Multiply by $(4x + 5)$:

$8x^3 + 10x^2 - 12x^2 - 15x - 36x - 45$

$= 8x^3 - 2x^2 - 51x - 45$.

3 marks total
First product
Distribute all terms
Collect coefficients

.....

(3)

(Total for Question 12 is 6 marks)

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